

Question	Answer	Marks
5(a)	region (of space) where a force acts on a (stationary) charge	B1
5(b)	$E = F / Q$	B1
	$F = ma$ and (so) $a = \frac{Eq}{m}$	A1
5(c)(i)	protons = 96	A1
	neutrons = 148	A1
5(c)(ii)	mass-energy is conserved/mass change is 'seen' as energy	B1
	energy released as gamma (radiation)/KE of α /KE of Pu	B1
5(c)(iii)	ratio = $\frac{2}{4} \times \frac{240}{94}$ or ratio = $\frac{2 \times 1.60 \times 10^{-19}}{4 \times 1.66 \times 10^{-27}} \times \frac{240 \times 1.66 \times 10^{-27}}{94 \times 1.60 \times 10^{-19}}$	C1
	ratio = 1.3	A1

9702/21

Cambridge International AS/A Level – Mark Scheme

October/November 2018

PUBLISHED

Question	Answer	Marks
$\sigma(a)$	sum of σ in $f(a)$ equal to sum of σ in $d(a)$	M1